

Probabilistic modelling of exposure doses and implications for health risk characterization: glycoalkaloids from potatoes.



Potatoes are a source of glycoalkaloids (GAs) represented primarily by alpha-solanine and alpha-chaconine (about 95%). Content of GAs in tubers is usually 10-100 mg/kg and maximum levels do not exceed 200 mg/kg. GAs can be hazardous for human health. Poisoning involve gastrointestinal ailments and neurological symptoms. A single intake of >1-3 mg/kg b.w. is considered a critical effect dose (CED). Probabilistic modelling of acute and chronic (usual) exposure to GAs was performed in the Czech Republic, Sweden and The Netherlands. National databases on individual consumption of foods, data on concentration of GAs in tubers (439 Czech and Swedish results) and processing factors were used for modelling. Results concluded that potatoes currently available at the European market may lead to acute intakes >1 mg GAs/kg b.w./day for upper tail of the intake distribution (0.01% of population) in all three countries. 50 mg GAs/kg raw unpeeled tubers ensures that at least 99.99% of the population does not exceed the CED. Estimated chronic (usual) intake in participating countries was 0.25, 0.29 and 0.56 mg/kg b.w./day (97.5% upper confidence limit). It remains unclear if the incidence of GAs poisoning is underreported or if assumptions are the worst case for extremely sensitive persons.